

model with shortlist softmax trained for 400k minibatch updates (65.26). Pointer softmax converges faster than the model using the shortlist softmax, because the targets provided to the pointer softmax also acts like guiding hints to the attention.

6 Conclusion

In this paper, we propose a simple extension to the traditional soft attention-based shortlist softmax by using pointers over the input sequence. We show that the whole model can be trained jointly with single objective function. We observe noticeable improvements over the baselines on machine translation and summarization tasks by using pointer softmax. By doing a very simple modification over the NMT, our model is able to generalize to the unseen words and can deal with rare-words more efficiently. For the summarization task on Gigaword dataset, the pointer softmax was able to improve the results even when it is used together with the large-vocabulary trick. In the case of neural machine translation, we observed that the training with the pointer softmax is also improved the convergence speed of the model as well. For French to English machine translation on Europarl corpora, we observe that using the pointer softmax can also improve the training convergence of the model.

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